

ENGINEERING PORTFOLIO

Marcus Gafka

> Passionate engineer, designer, roboticist, and hobbyist

— DEKA Research and NASA JSC Intern

— Worcester Polytechnic Institute Robotics and Mechanical Engineering student

— First Robotics Competition Team 190 Mentor and Drive Coach

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About Me

I grew up in Houston's Clear Lake area, just down the road from NASA's Johnson Space Center. When rockets are part of the local scenery, building machines that move starts to feel like the obvious thing to do. That curiosity never really left, and it's what led me to robotics.

Robotics eventually took me halfway across the country to New England. I came to WPI on its FRC scholarship, where I'm pursuing degrees in Robotics Engineering and Mechanical Engineering and will graduate in May 2027.

I'm interested in the part of robotics where mechanical systems, controls, software, and real-world uncertainty all collide. I like building machines that have to sense their environment, make decisions, move precisely, and keep working when things don't go exactly as planned.

After graduation, I'm looking for a full-time role in robotics or controls engineering. If you're tackling problems that don't have easy answers, building something that has to sense, plan, decide, act, move, fail, recalibrate, recover, replan, and try again, I'd love to talk.

Work Experience

Robotics and controls internships in industry and at NASA.

Controls Engineering Intern: DEKA Research & Development

Summer 2026 · Controls Engineering Intern · Manchester, NH · marcusgafka.com/#/work/deka

Closed-loop stepper motor control, frequency-response characterization, and sensor filtering for prototype testing.

Arduino | Python | Kalman Filtering | Controls

Overview

Summer 2026 controls engineering internship at DEKA Research & Development in Manchester, NH.

> **Why this page is brief:** my work at DEKA is covered by a non-disclosure agreement, so I can't share photos, designs, data, or details about the products involved. Below is what I'm able to share publicly.



What I Did

- Developed an Arduino-based platform with magnetic encoder feedback to validate closed-loop stepper motor control for testing prototypes
- Developed Python GUI-controlled software and automated Bode plot generation from real-time data to characterize the frequency response of a nonlinear system
- Diagnosed sensor limitations including noise, resolution, and system latency, then improved control performance using tailored Kalman filters and hysteresis logic

Robotics Academy Intern: NASA Johnson Space Center

Summers 2023–2025 · Robotics Academy Intern / University Intern · Houston, TX · marcusgafka.com/#/work/nasa-jsc

Space Exploration Vehicle maintenance, a new door latch design, and dynamic trophies for the Space City VEX event, across three summers.

Mechanical Design | Fabrication | Sheet Metal | 3D Printing

Overview

Three summers (2023, 2024, 2025) at NASA Johnson Space Center in Houston, TX: first as a NASA Robotics Academy High School Intern, then as a Robotics Academy University Intern. Alongside the two projects below, I worked on:

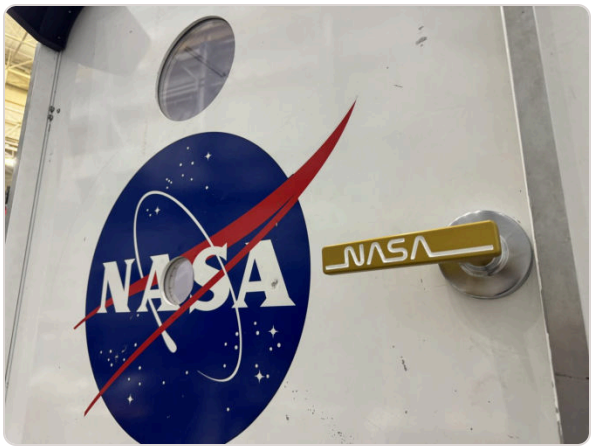
- **Lunar Terrain Vehicle (LTV) Ground Test Unit (GTU):** assembled critical components of the new rover, including the chassis, electronics housings, human operator interfaces, and differential swerve wheel modules, and later led construction of a high-fidelity mock-up of the vehicle's aft section for the tool carrier assembly
- **Space Exploration Vehicle (SEV) GTU:** maintained the vehicle for demonstrations and training
- **Harmonic gearbox demonstrator:** led the design of a hand-powered harmonic gearbox that shows how this compact, high-ratio drive works and why it's valuable for space applications

Space Exploration Vehicle Door Latch

Managed the design, integration, and schedule for a new door handle and latching mechanism on the SEV GTU. The design puts a large emphasis on ruggedness so the vehicle can be tested at the desert field-test site, and the handle is sized for astronauts operating it in full suits during mock missions.



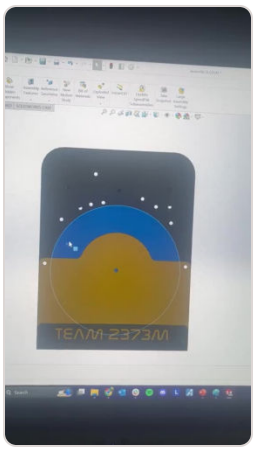
The Space Exploration Vehicle



The new SEV door handle

Trophy Development

Led the design and manufacturing of dynamic trophies for the Space City VEX Signature Event.



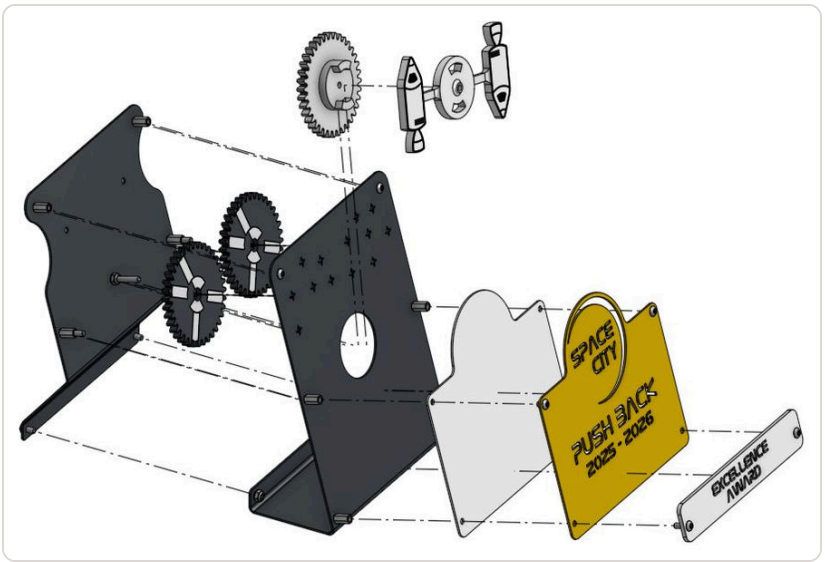
Video still · *Motion study in CAD*



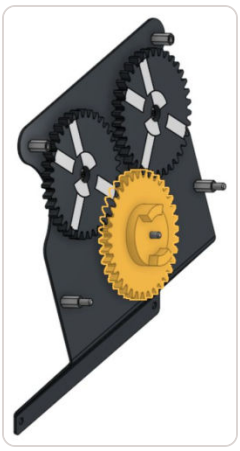
Early prototype

Trophies are designed to be fabricated in batches, with a new set manufactured season after season:

- Waterjet-cut and bent aluminum stand with front, back, and logo plates
- 3D-printed gear train and rocket mounted on the front plate
- Polycarbonate backing and award plates
- Assembled entirely with standard VEX screws, standoffs, and nuts, with an ordering sheet sized for a run of 40 trophies
- Only the season plate changes each year, from High Stakes (2024–25) to Push Back (2025–26)
- Documented in a 20-page build manual: full parts list, ordering sheet, and a 21-step assembly guide
- Featured in FUN Robotics Network's video Incredible Trophies | Space City Signature



Exploded view of the final design



Gear train



Rocket



Finished trophy



Video still · *The finished trophy in motion*



Push Back 2025–26 Build and Excellence Awards



All 44 trophies, ready for the event

Worcester Polytechnic Institute

Full tuition
Sole recipient of WPI's merit-based FRC scholarship

Double major
B.S. Robotics Engineering & B.S. Mechanical Engineering

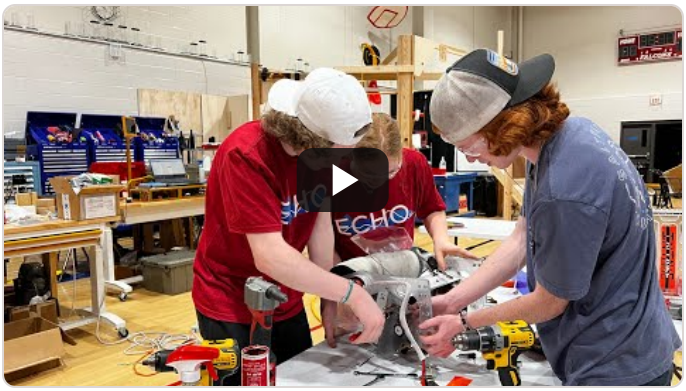
3.94 GPA
Graduating May 2027

Venice, Italy
Study abroad: Buildings of Venice IQP

New Member Officer
Rho Beta Epsilon Robotics Honor Society

WPI FRC Scholarship

Robotics is what brought me to WPI. As a high school senior in 2023, I applied for WPI's robotics scholarships for students from both the FIRST Robotics Competition and the VEX Robotics Competition, submitting a video for each. I was selected as the **2023 winner of WPI's full-tuition, merit-based FRC scholarship**.



youtube.com/watch?v=nL_hcSbydXM

My FRC scholarship submission (2023)



youtube.com/watch?v=kTsBi09B_vU

My VEX scholarship submission (2023)

ROBOTICS HONOR SOCIETY (RHO BETA EPSILON)

Rho Beta Epsilon: New Member Officer

September 2025 – present · **Elected officer** · marcusgafka.com/#/wpi/rho-beta-epsilon

Elected New Member Officer of WPI's Robotics Engineering honor society: leading recruitment, interviews, and induction of new members.

Leadership | Interviewing | Recruitment

Overview



Rho Beta Epsilon is WPI's Robotics Engineering honor society (Alpha Chapter). Its mission is *"to encourage and recognize exemplary robotics students while aiding in their development both personally and professionally through exceptional education, leadership, and passion for robotics; and to enable the advancement of the robotics engineering community."* Members support the RBE department through workshops, lab help hours, and events each term.

Membership

Membership is selective and open to RBE majors and minors in good academic standing. Undergraduates need three full semesters at WPI and two RBE courses at the 2000 level or above. Each application cycle, at the start of A-Term and C-Term, applicants:

- Solve challenging technical questions
- Submit a résumé and statement of purpose
- Present their accomplishments in a professional interview

My Role: New Member Officer

In September 2025 I was elected New Member Officer, responsible for bringing in each new class of members:

- Lead recruitment and induction for prospective members
- Develop the technical questions, scoring rubrics, and interview process
- Coordinate applicant interviews and new member evaluations

LINKS

Rho Beta Epsilon ↗

COURSEWORK · CURRENT

RBE4815 – Industrial Robotics

B-Term 2026 · Lab team of 5 · marcusgafka.com/#/wpi/rbe4815

Kinematics of 6-DOF industrial arms: forward and inverse kinematics with a spherical wrist, Jacobians, singularities, and workspaces, with labs on a UR5e and a FANUC delta robot.

Kinematics | DH Parameters | Inverse Kinematics | Jacobians | UR5e | FANUC

Overview

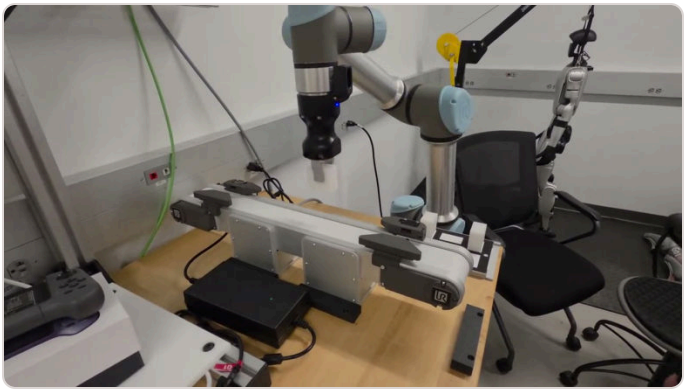
In this class I learned the math behind the serial and parallel robots used in factories: finding where a 6-axis arm's end effector is, solving for the joint angles that get it somewhere, and spotting where the robot loses the ability to move. In the labs, I put it into practice on real industrial robots.

Concepts

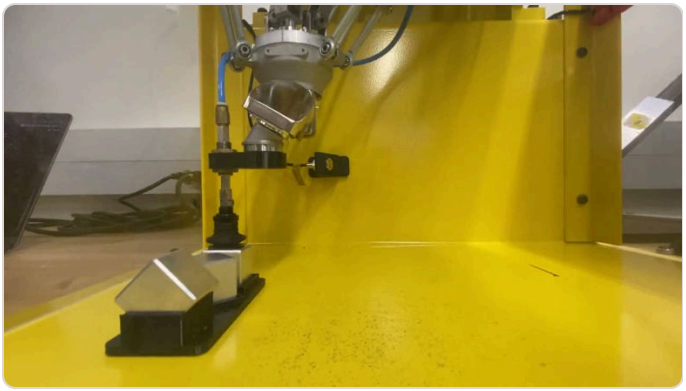
- **Robot types and workspaces:** serial arms (articulated, SCARA, spherical, cylindrical, Cartesian) versus parallel robots, and the volume each one can actually reach
- **Forward kinematics:** Denavit–Hartenberg parameters and chained homogeneous transforms from the base to the tool; a spherical wrist is just a ZYZ Euler rotation
- **Inverse kinematics by decoupling:** solve for the wrist center's position with the first three joints (geometry and the law of cosines, giving elbow-up and elbow-down solutions), then for orientation with the wrist's Euler angles
- **Velocity kinematics:** the Jacobian maps joint velocities to tool velocities, $\dot{\mathbf{p}} = J(\mathbf{q})\,\dot{\mathbf{q}}$, and inverting it gives the joint speeds for a desired tool motion
- **Singularities:** when the Jacobian loses rank, the arm loses a direction of motion and nearby joint speeds blow up; at the edge of the workspace, with two axes lined up, or when the wrist's first and last axes align
- **Numerical IK and the analytical Jacobian:** iterative Jacobian-based solvers for arms with no closed-form answer, and relating angular velocity to Euler-angle rates

Labs

In a team of five, we programmed industrial robots in the lab: a Universal Robots **UR5e** collaborative arm and a **FANUC** delta robot.



Video still · *UR5e picking and placing blocks (2× speed)*



Video still · *FANUC delta robot*

COURSEWORK · CURRENT

MQP – Major Qualifying Project – FloorJet, a Mobile Floor-Painting Robot

2026–2027 · Senior capstone · Team project · marcusgafka.com/#/wpi/mqp

A mobile robot that paints images onto floors: image-to-path software, a spray-painting nozzle, and precise localization. Year-long senior capstone, in progress.

Conceptual Design | Path Planning | Image Processing | Axiomatic Design | Simulation

> **Work in progress.** FloorJet is my year-long Major Qualifying Project (senior capstone), August 2026 – spring 2027. So far: research, requirements, and system design; next come detailed design, prototyping, testing, and a final demonstration.

Overview

FloorJet is a mobile robot that paints images directly onto floors: logos on gym floors, wayfinding lines, hazard markings, booth footprints. A user uploads an image and places it in the room; software turns it into paint strokes and a driving path, and the robot drives and sprays to reproduce it.

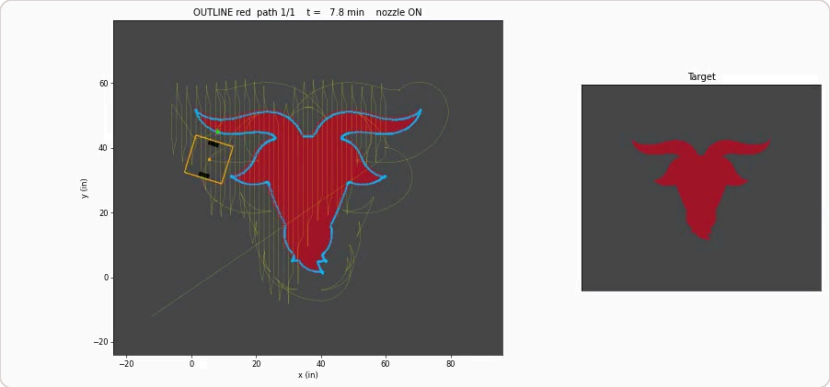
Design Direction

Research into floor-marking robots, robotic painting, and commercial systems shaped the requirements:

- A compact **differential-drive** robot with external motion-capture localization, holding the nozzle within ±0.25 in
- An adjustable spray footprint (about 1–3 in), **two or more pre-mixed colors** with no mixing on the robot, and one-pass coverage
- No fixed canvas size, and software that **never drives over wet paint**

Simulation

Each shape is painted in two passes: **raster** strokes fill it, then **vector** strokes trace the outline for clean edges. Before building hardware, we simulated an airbrush robot doing exactly that to study how spray size affects print quality.



Video still · *Simulated airbrush robot painting the image*

COURSEWORK · COMPLETED

RBE4540 – Vision-Based Robotic Manipulation

A-Term 2026 · marcusgafka.com/#/wpi/rbe4540

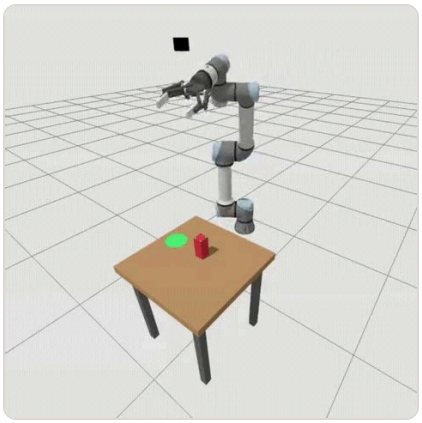
The theory behind robotic grasping: grasp matrices, grasp quality metrics, vision-based feature detection, and visual servoing, in ROS 2 and Gazebo.

Grasp Analysis | ROS 2 | Gazebo | OpenCV | Visual Servoing | Python

Overview

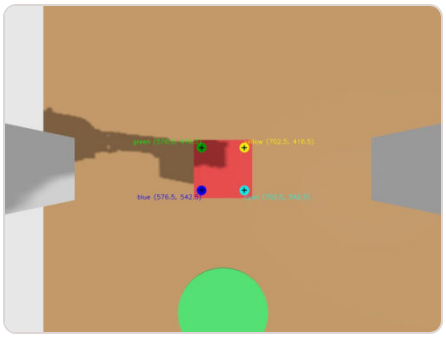
In this class I learned the theory behind how robots grasp objects reliably and the vision that guides them, with hands-on work in **ROS 2**, a **Gazebo** simulation of a robot arm with a palm camera, and **OpenCV**.

Concepts

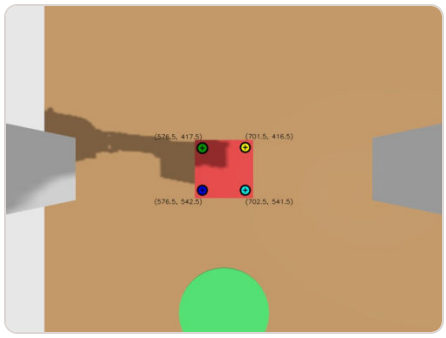


Video still · *Vision-guided pick and place in simulation*

- **Grasp taxonomies:** classifying human grasps (power vs. precision, which fingers oppose, thumb position) and using object properties to choose a robot gripper
- **The grasp matrix:** a linear map from the forces at each fingertip to the net force and torque on the object, built from each contact's position and orientation and a contact model (frictionless, hard finger, or soft finger)
- **Grasp quality:** the singular values of the grasp matrix describe how well a grasp resists disturbances; different metrics (weakest direction, overall strength, or how evenly balanced it is) can disagree on the "best" grasp
- **Vision features:** color thresholding, edge, corner, and circle detection to locate an object from the palm camera
- **Visual servoing:** closing the control loop on the image itself, turning the error between where features are and where they should be into camera velocities through an image Jacobian



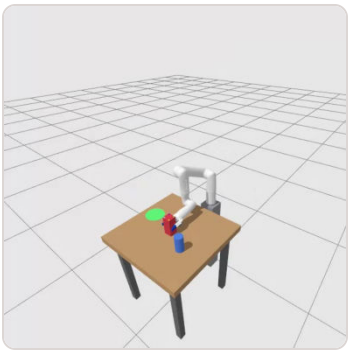
Color thresholding: marker centroids



Hough circle detection

Final Project: Grasping from Point Clouds

In a group, we built a pipeline that finds and picks up objects using only a wrist-mounted depth camera, first in Gazebo and then on a real **Kinova Gen3 lite** with a RealSense D405.



Video still · *Planned grasp in simulation*

- **Point clouds:** captured from two camera poses, merged in the robot's base frame, and downsampled
- **Segmentation:** removed the table plane and split what remained into individual objects
- **Grasp planning:** searched contact-point pairs for a stable two-finger grasp inside the friction cone, then chose an approach direction from the object's principal axes (PCA)
- **Execution:** MoveIt moves to a pre-grasp pose, approaches in a straight line, closes the gripper, and lifts



Video still · *Real robot: cube (5× speed)*



Video still · *Real robot: rectangular block (5× speed)*

COURSEWORK · COMPLETED

RBE4701 – Artificial Intelligence for Robotics

A-Term 2026 · marcusgafka.com/#/wpi/rbe4701

Search, adversarial reasoning, Markov decision processes, and reinforcement learning, applied to a Bomberman AI that escapes monsters with A* and value iteration.

Python | A* | Minimax | MDPs | Value Iteration | Reinforcement Learning

Overview

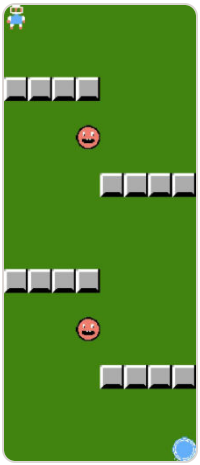
In this class I learned the core tools a robot uses to decide what to do, implementing each one in **Python**:

- **Search:** breadth-first, uniform-cost, and A*, where an admissible heuristic that never overestimates guarantees the cheapest path
- **Adversarial search:** minimax, alpha-beta pruning to skip branches that can't matter, and expectimax when the opponent is random
- **Local search:** hill climbing and simulated annealing, which occasionally accepts worse moves to escape local optima
- **Markov decision processes:** modeling uncertain outcomes with transitions, rewards, and a discount factor, and solving them with value and policy iteration
- **Reinforcement learning:** learning values from experience when the model is unknown, either by estimating the model or updating values directly (temporal-difference learning)

Applied: A Bomberman AI

In a team of three, we built one AI to escape five increasingly dangerous versions of a Bomberman level, from an empty map to one with a random monster and an aggressive one that chases you.

- **A*** plans the shortest route to the exit whenever no monster is nearby
- When a monster gets close, the character switches to **value iteration** on an MDP over the character's and monster's positions, rewarding escape, heavily penalizing the monster's attack range, and modeling the monster as chasing or wandering
- Because the walls never change, the value table is computed once and cached, restricted to nearby character/monster pairs to cut the state space by about 38%
- A headless test harness ran thousands of games without rendering, wired into pytest and CI



Video still · *Escaping both monsters (variant 5)*

Variant	Escape rate (1,000 games each)
1. Alone	100.0%
2. Random monster	99.7%
3. Self-preserving monster	93.8%
4. Aggressive monster	66.4%
5. Random + aggressive monsters	81.7%

One algorithm handled all five variants, with an overall 86.3% success rate, without ever using bombs.

COURSEWORK · COMPLETED

IQP – Interactive Qualifying Project

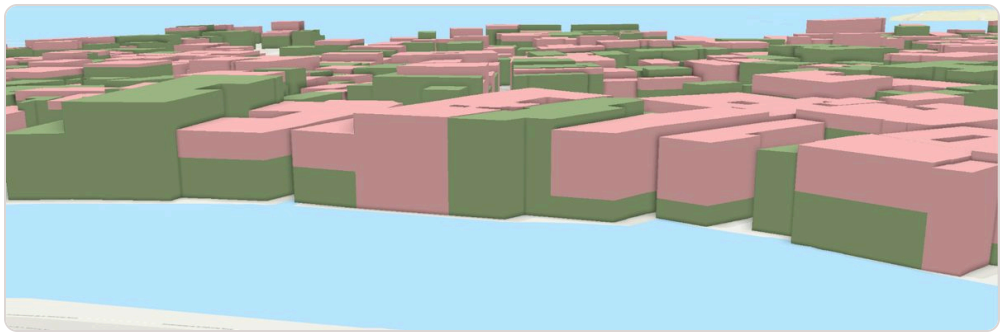
B-Term 2025 · Team of 4 · Venice, Italy · marcusgafka.com/#/wpi/iqp

The Buildings of Venice: Venice's first building-by-building estimate of residential, tourist, and vacant units as well as overall population, built from city data and fieldwork on site.

ArcGIS Pro | Survey123 | Python | GIS | Linear Regression | Fieldwork

Overview

The Buildings of Venice was my Interactive Qualifying Project, completed on site in Venice, Italy, in partnership with **SerenDPT** and the Venice Project Center. Working in a team of four (Robotics & Mechanical, Mechanical, Computer Science, and Chemical Engineering majors), we built **Venice's first building-by-building estimate** of residential units, tourist units, vacant units, and population for the historic city center.



Estimated residential (green) vs. non-residential (pink) share of each building's height

The Problem

Venice's geography keeps it from expanding, so its existing buildings have to house full-time residents, second-home owners, and a year-round flood of tourists. As short-term rentals and hotels replace residential housing, availability shrinks, prices rise, and the city keeps losing residents.

To address depopulation, policymakers need to understand housing **at the building level**, but population data only existed per census tract, and a tract can contain anywhere from 1 to 100 buildings.

Goals

- Organize SerenDPT's scattered city datasets into a single, usable GIS dataset
- Design a repeatable fieldwork method to verify the data on the ground
- Estimate floors, residential / non-residential / vacant units, and residents for every building
- Publish the methods and datasets so future teams and the city can update them

Building a Unified GIS Dataset

The city's building data was often mislabeled or outdated, with hundreds of undocumented fields that had to be translated from Italian and cross-referenced. Most critically, the gutter-height field was unusable for **nearly 5,000 buildings**, about a third of the historic center.

- Spatially joined addresses, census tracts, and water meters to buildings for the first time in ArcGIS Pro, so each meter links to an address, each address to a building, and each building to a tract
- Stripped irrelevant fields and renamed the rest so the layers are readable without insider knowledge

- Created a new **alias system** to replace the city's arbitrary numeric IDs: each alias encodes the sestiere, island, census tract, and building number, snaking the city from northwest to southeast, so a building's location can be read from its code alone

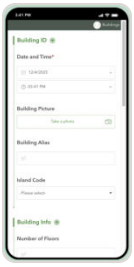


Building aliases on one island

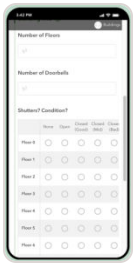
Fieldwork in Venice

With **15,485 buildings** and eight weeks, visiting every building was impossible, so we designed an organized, efficient process that future Venice IQP teams can repeat.

- Built a **Survey123** building survey capturing photos, alias and island code, floor count, doorbell count, and shutter condition per floor
- Doorbells validated unit counts; closed, worn shutters validated vacancy estimates
- Walked islands with alias maps and an alias-to-address spreadsheet, so we always knew exactly which building we were surveying, even when neighbors shared a gutter line and color
- Ran nighttime fieldwork measuring building heights, primarily in San Polo



Survey123 building survey

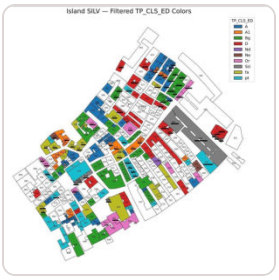


Floors, doorbells, and shutters

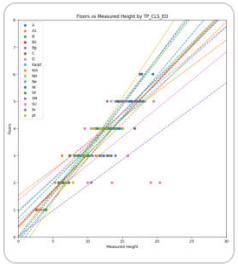
Estimating Floors from Building Type

Venetian buildings vary enormously, so no single rule fits them all. We identified the most common architectural classifications and measured heights and floor counts for about **15 buildings in each of 13 types (239 data points)**.

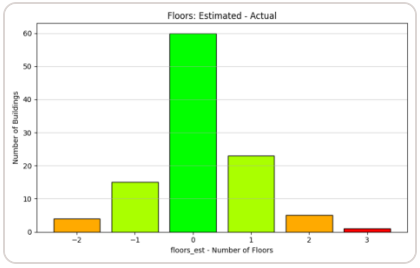
- Fit a linear regression per building type to get its average floor height
- Estimated floors for any building as its height divided by its type's floor height
- For the ~5,000 buildings with no usable height, estimated height from the **5 nearest buildings of the same type**
- Implemented the full pipeline in **Python**, producing a floor estimate for every building in Venice



Buildings colored by architectural type



Floors vs. measured height, per building type

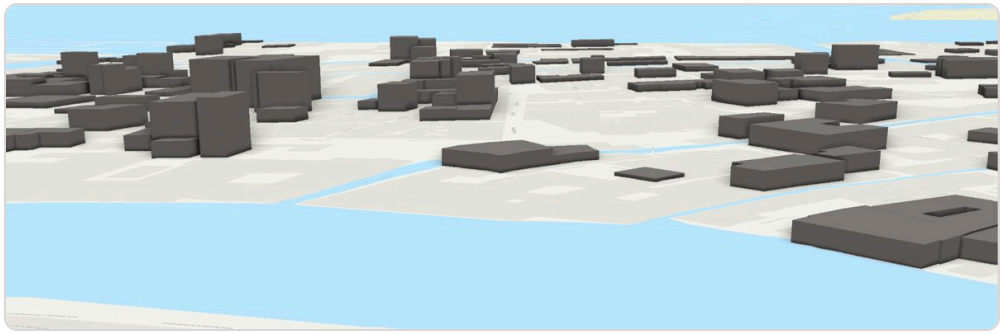


Estimated minus actual floors

Occupancy Model

The final model combines the 2021 census, water-meter records, building classifications, measured and estimated heights, short-term rental locations, and non-residential buildings:

- **Units:** each water meter generally corresponds to one unit, and residents, non-residents, and businesses pay different water rates, so meters also reveal how each unit is used
- **Vacancy:** residential units with zero water use in 2024 were marked vacant, and fieldwork showed a statistically significant link between closed shutters, zero-use meters, and vacancy
- **Population:** each tract's census population was distributed across its buildings by residential unit count and footprint, excluding hotels and tourist accommodations



Estimated vacant share of each building's height

Results

- Delivered **Venice's first per-building population and unit estimation model**, updatable whenever new census or water-meter data is released and replicable in other cities facing over-tourism
- Showed that San Marco, San Polo, and Dorsoduro have lost the most residential use to tourism, while Santa Croce, Cannaregio, and Castello remain far more residential
- Published graphs, StoryMaps, ArcGIS layers, and datasets in a public repository for city officials, residents, and future IQP teams
- Documented every assumption so the estimates can be checked, refined, and extended

My Role

On site in Venice, I did hands-on field data collection: walking islands with our Survey123 building survey to record floors, doorbells, and shutter condition building by building, and taking nighttime building-height measurements. I also co-wrote the abstract, introduction, background on Venice's infrastructure history and geography, the methodology for building the unified GIS dataset and field data collection, ethical considerations, and the findings on the unified dataset and residential occupancy, and I helped edit every chapter of the final report.

Tools: ArcGIS Pro, ArcGIS Survey123, Python, spatial joins, linear regression

Advisors: Judith Nitsch and Luis Vidali · **Sponsor:** Dr. Fabio Carrera, SerenDPT

COURSEWORK · COMPLETED

RBE3002 – Unified Robotics IV: Navigation

D-Term 2025 · Team of 4 · marcusgafka.com/#/wpi/rbe3002

Programmed a TurtleBot3 with ROS to explore, map, and localize in an unknown arena. Finished in 6 of 15 allotted minutes.

ROS | Python | Linux | Git | Path Planning

Overview

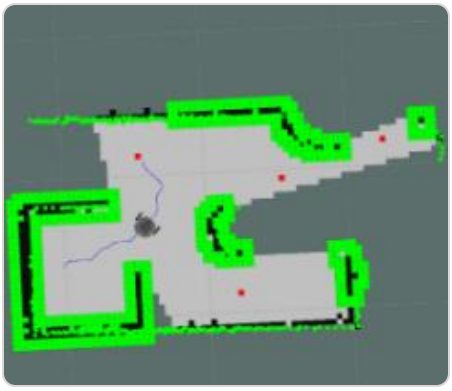
In a team of four, programmed a TurtleBot3 (a differential-drive robot with a LIDAR sensor) using ROS to autonomously explore and map an unknown arena, then localize and navigate within it. The team built a custom path planner, motion controller, and communication protocol, using Linux and Git to manage the code.

Phases 1 & 2: Exploration and Mapping

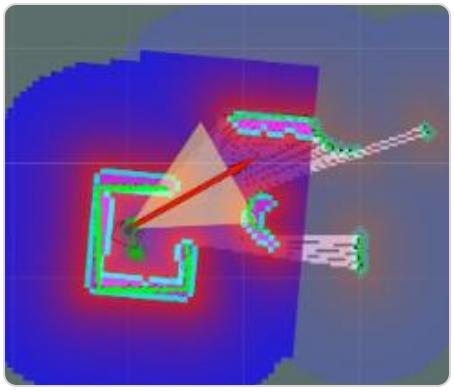
- Detect obstacles and create a keep-out zone (C-space)
- Identify frontiers: unexplored areas at the edge of the map
- Intelligently select a frontier and plan an efficient path
- Follow the path accurately while maintaining speed
- Save the mapped environment when exploration is done

Phase 3: Localization

- Place the robot at a random location
- Run a calibration maneuver while collecting sensor data
- Determine the robot's position in the saved map
- Quickly traverse the mapped area



C-space and frontiers (red)



Localizing in the saved map



Final demo

Results

- Finished in **6 minutes** against a 15-minute time limit
- Stayed centered in narrow hallways while continuously updating its location and map
- Received an **A+** after the preliminary design review, live demo, individual code review, and project report

LINKS

Lab Report ↗

COURSEWORK · COMPLETED

RBE3001 – Unified Robotics III: Manipulation

C-Term 2025 · Team of 4 · marcusgafka.com/#/wpi/rbe3001

Forward and inverse kinematics for a 4-DOF arm that uses a camera to find and sort balls by color.

MATLAB | Kinematics | Computer Vision

Overview

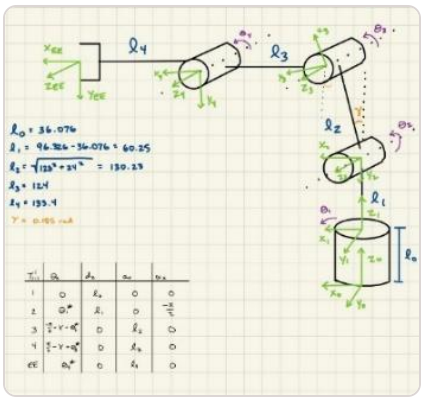
In a team of four, programmed a 4-DOF robot arm in MATLAB to move smoothly and use camera data to detect objects and sort them into compartments by color, implementing forward and inverse kinematics with DH transformations.



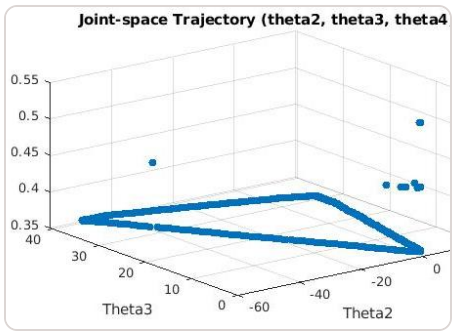
Picking up a green ball

Kinematics

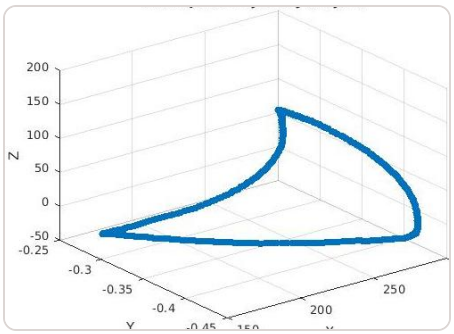
- Initial geometric analysis and home position assignment
- DH table describing the transformation between consecutive joints
- Established the math behind all arm motion



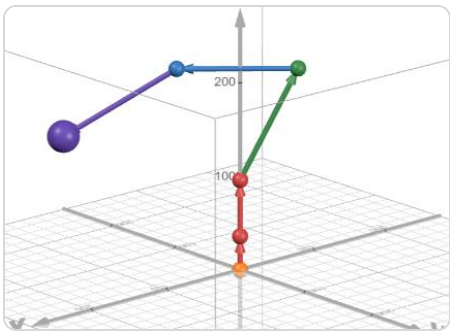
Arm coordinate systems



Joint-space trajectory



Task-space trajectory



Desmos 3D arm model

Results

- Received an **A+** after the live demo and project video
- Built a Desmos 3D calculator and visualization tool to simulate arm motion and validate the equations of motion

LINKS

Project Video ↗ | Lab Report ↗ | Desmos 3D Model ↗

COURSEWORK · COMPLETED

RBE2002 – Unified Robotics II: Sensing and Perception in Robotics

B-Term 2024 · Team of 3 · marcusgafka.com/#/wpi/rbe2002

PID control, line following, AprilTag tracking, and state machines on a Pololu Romi. Worked on the first attempt.

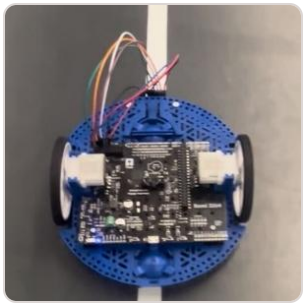
C++ | PID Control | OpenMV | State Machines

Overview

In a team of three, programmed a robot to autonomously complete a series of challenges: PID control, state machines, line following, AprilTag detection and tracking, strain gauge measurement, hysteresis, gyroscope optimization, and more. Everything ran in C++ on a Pololu Romi, culminating in an autonomous trash collection demo.

Line Following

- PID controller for wheel velocities
- Line sensor array to detect surface color
- Tuned PID controller for line following
- Custom logic for intersection detection
- Filtered gyroscope bias and drift for precise turns



Following a white tape line

Final Robot

- Differential-drive chassis with motors and encoders
- Romi board with IR receiver, gyroscope, and more
- Array of six optical sensors mounted close to the ground
- Servo-actuated over-center lifting mechanism
- OpenMV camera mounted below the lifter



All final systems attached

Results

- Worked on the first attempt: collected 3 buckets and delivered them to the drop-off zone
- Finished with time to spare for extra credit challenges, earning an **A+**
- Implemented a state machine with sub-states

LINKS

Development Video ↗

COURSEWORK · COMPLETED

RBE1001 – Introduction to Robotics

B-Term 2023 · Team of 3 · marcusgafka.com/#/wpi/rbe1001

Designed, built, and programmed a VEX robot in Python that won the class-wide competition.

Python | VEX | Sensors | State Machines

Overview

In a team of three first-year students, built a VEX robot to compete in a class-wide competition, backing up the design with engineering calculations. We used sensors and state machines for reliable autonomous routines.

Robot

- Conveyor belt that carries 4 balls at a time
- Grabber with a rubber tip for tipping the bucket
- Fast four-motor drivetrain for speed and torque
- High-traction wheels for climbing the steep ramp
- Ultrasonic, line, camera, and gyroscope sensors



Our completed robot



Scoring through the blue goal



Tilting the bucket to get the yellow ball

Results

Scored **100** on the project by:

- Winning the class-wide competition
- Completing the extra credit challenges
- Writing a comprehensive, detailed report

LINKS

Lab Report ↗ | Competition Footage ↗

COURSEWORK · COMPLETED

ME3310 – Kinematics of Mechanisms

C-Term 2025 · Team of 4 · marcusgafka.com/#/wpi/me3310

A foot-pedal-actuated six-bar linkage, prototyped at 1:3 scale for under $30 and pitched Shark Tank style.

Linkage Synthesis | CAD | Laser Cutting | 3D Printing

Overview

In a team of four, synthesized a mechanism that solves a real-world problem, built a working prototype, and gave a Shark Tank–style pitch to the professor.

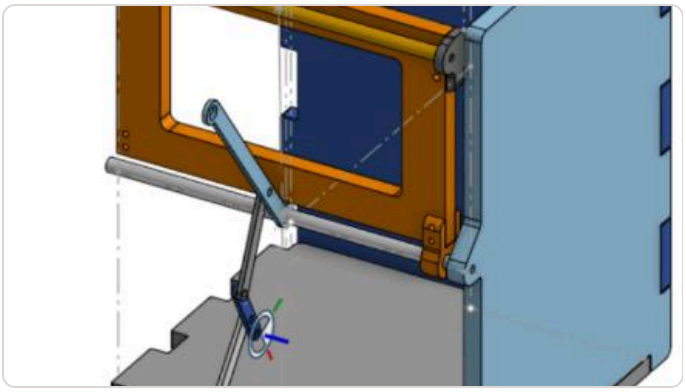
Goals

- Create a unique mechanism with more than 4 links and 4 joints
- Develop the CAD and build a fully functional prototype for under $30

Final Design: Oven Door Opener



1:3 scale prototype



CAD

- Foot-pedal-actuated oven door opener
- Six-bar linkage in a Watt arrangement
- Laser-cut wood frame, acrylic links, and cardstock stencils
- 3D-printed PLA pedal and connectors
- Iterated on screw joints for minimal slop and friction

Results

- Prototype was fully functional and demonstrated live
- Well received by the professor and classmates
- Received an **A+** after the product pitch and report

LINKS

Final Report ↗

Competitive Robotics

The competition robots I've designed, built, driven, and mentored, newest first: from mentoring FIRST Robotics Competition Team 190 at WPI back to captaining FRC Team 118 in high school.

FRC Team 190 · Mentor and Drive Coach

Since 2023 I've been a college mentor for FIRST Robotics Competition Team 190 at WPI, putting in about 400 hours a year plus travel to competitions. I design and build mechanisms side by side with high school students, and I'm currently the team's drive coach, responsible for match strategy, coordinating with alliance partners, and running the team on the field.

The Blue Alliance ↗ YouTube ↗



Team 190 and our alliance with the winner banners

FRC TEAM 190 · SEASON ROBOTS

Doom Spiral / Turnover

2026 Season · Prototyping Lead · V2 Hopper Lead · Drive Coach · marcusgafka.com/#/robotics/doom-spiral-turnover

2026 in-season robots. Held many roles at once: Prototyping Lead, V2 Hopper Lead, and Drive Coach.

Prototyping | CAD | Mechanism Design | Leadership | Strategy

Overview

In the 2026 season I held many roles on the team at the same time:

- **Prototyping Lead:** managed concurrent prototypes of potential robot subsystems
- **Hopper Lead:** as mentor roles shifted over the season, led the design of the V2 robot's complex hopper subsystem
- **Drive Coach:** continued running the team on the field at competitions

Prototyping

- Managed concurrent prototypes of various potential robot subsystems
- Helped with the intake, many prototype hopper mechanisms, and full robot layout concepts



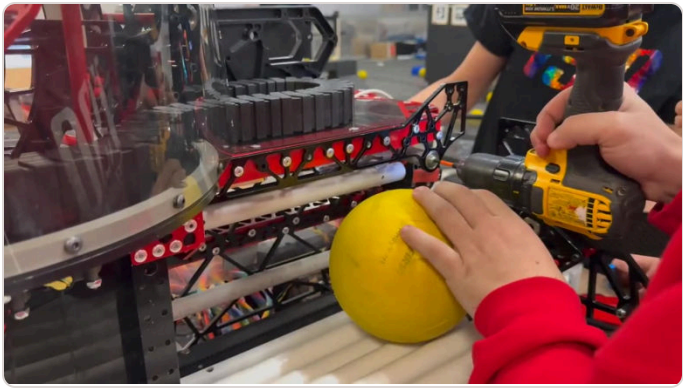
Early hopper prototypes: turning a collection of balls into a single stream

Hopper

- Led the design of the V2 robot's complex hopper subsystem
- Throughput of up to 10 balls per second



V2 hopper system CAD



Video still · *The first ball going through*



Video still · *Pre-Worlds driver practice with the V2 robot, fully tuned for maximum throughput*

LINKS

Reveal Video ↗ | Worlds Match E2 ↗

FRC TEAM 190 · SEASON ROBOTS

Stack Up / Redundancy

2025 Season · Funnel & Stick subsystem lead · Drive Coach · marcusgafka.com/#/robotics/stack-up-redundancy

2025 in-season robots. Led the Funnel and Stick subsystems; 3x district event winner and ranked 1st in New England.

CAD | Mechanism Design | Strategy

Overview

Contributed to high-level robot strategy and architecture, led the **Funnel** and **Stick** subsystems, trained the new drive team, and helped the team succeed at competitions as Drive Coach.

Goals

- Build two high-quality robots and qualify for the World Championship
- **Funnel** as wide as possible with robust game piece acquisition
- **Funnel** must clear space for the climber at the end of the match
- **Stick** passively manipulates the game piece

Funnel

- Motor-driven opposing wheels and wings
- Retro-reflective tape and sensor to detect the game piece
- Mounted to angled crossbeams for the elevator
- Wings close to intake the game piece, then open to clear space for the climber



Funnel CAD

Stick

- Added to the completed robot without an additional motor
- Extreme weight reduction to stay under 1 lb
- One-way bearing enables a passive four-bar linkage
- Surgical tubing spring ensures two discrete positions
- Custom rubber wheel for game piece manipulation



Stick CAD

Results

- Competed at 7 events: **3x District Event Winner**, **New England Championship Division Winner**
- **Ranked 1st in New England** in district points
- Received the Quality, Industrial Design, and Creativity Awards
- One of the best intake systems at the 2025 FRC World Championship

LINKS

The Blue Alliance ↗ | Development Video ↗ | Reveal Video ↗ | Worlds Match Q106 ↗

FRC TEAM 190 · SEASON ROBOTS

Snapback

2024 Season · Intake & Centralizer lead · Strategy Lead · marcusgafka.com/#/robotics/snapback

2024 in-season robot. Led the Intake and Centralizer subsystems; World Championship Curie Division winner.

CAD | Pneumatics | Mechanism Design

Overview

Led the **Intake** and **Centralizer** subsystems from ideation and prototyping through design and manufacture, and served as Strategy Lead during competitions.



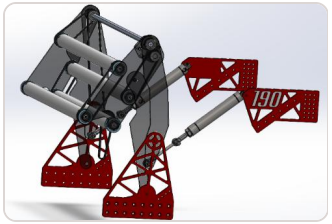


Goals

- Build a high-quality, competitive robot
- **Intake** must absorb impacts without breaking
- **Intake** must be full width with robust acquisition

Intake

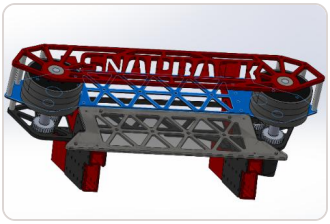
- Pneumatic-piston-powered four-bar linkage
- Rollers driven by belts from a motor mounted low in the chassis
- Makes full use of the allowed 18-inch extension
- Stows compactly above the Centralizer



Intake CAD

Centralizer

- Opposing wheels powered by a single motor
- Moves the game piece to the center of the robot
- Magnetic door cover for battery access



Centralizer CAD

Results

- Competed at 11 events: **2x District Event Finalist**, **World Championship Curie Division Winner**
- Ranked 32nd in New England in district points
- Quality and Excellence in Engineering Awards at district events
- Intake and Centralizer met their goals and consistently outperformed expectations

LINKS

The Blue Alliance ↗ | Intake Development Video ↗ | BattleCry Match E23 ↗

FRC TEAM 190 · OFF-SEASON

Funky

2024 Off-Season · Designer & builder, alongside students · marcusgafka.com/#/robotics/funky

A modular test-platform robot for prototyping software and mechanisms ahead of the 2025 season.

CAD | Fabrication | Electrical Integration

Overview

Brainstormed, designed, manufactured, assembled, and iterated on Funky side by side with students: a test platform for prototyping software and mechanical systems.

Goals

- Prepare Team 190 for the 2025 build season
- Enable early-season driver practice
- Focus on modularity to adapt the robot as needed

Build



During assembly



With added systems

Chassis

- Box tube and gusset construction with cross-members for robustness
- Mounting rails on either side for easy integration
- Designed to quickly add other systems

Added systems

- Electrical components fully integrated and operational
- Ballast to simulate the dynamics of the competition robot
- Upper body of the 2025 Kitbot for early practice
- Vision system so the software team could work on camera placement and field localization

Results

- Enabled early development and testing of both mechanical and software systems
- Gave the drive team weeks of extra driver practice
- Enabled experimentation with AprilTag detection, vision-based localization, and autonomous routines
- Provided a reliable, adaptable platform for subsystem prototyping, contributing directly to the 2025 robot's success
- Doubled as a defensive robot for practicing against aggressive opponents

FRC TEAM 190 · OFF-SEASON

Ember

2023 Off-Season · Project lead · marcusgafka.com/#/robotics/ember

An experimental swerve-drive robot with a ground cube intake, taken from idea to competing in three weeks.

Swerve Drive | CAD | Rapid Prototyping

Overview

Led the initial scoping, rapid design, and integrated assembly of Ember, an experimental swerve-drive chassis with a ground cube intake.

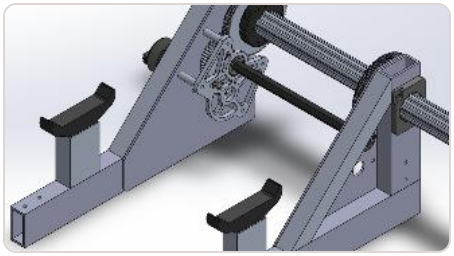
Goals

- Create an experimental swerve-drive robot for 190 to practice programming and driving
- Onboard a small group of college mentors
- Teach incoming students prototyping and fabrication
- Go from project start to functional robot in only **3 weeks**

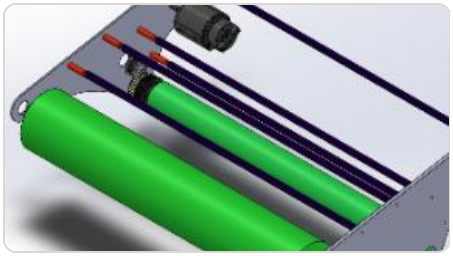
Build



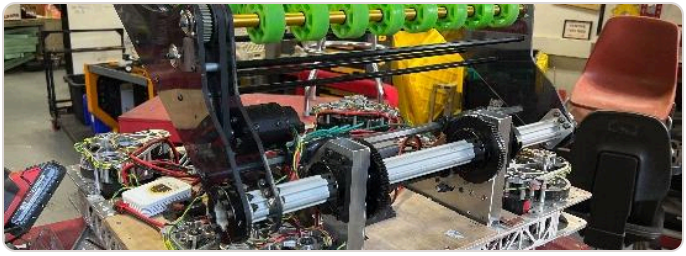
Initial intake prototyping



Early CAD of the pivot gearbox



Early CAD of the intake



Fully assembled and wired



Testing intaking and scoring cubes

Results

- Competed at the River Rage off-season event
- Let 190 transition seamlessly to swerve drive in later seasons
- Used in driver tryouts for the 2024 season

LINKS

Development Video ↗ | River Rage Match E6 ↗

HIGH SCHOOL · FRC TEAM 118 ROBONAUTS

FRC Team 118 Robonauts

2019–2023 · Team Captain & Drive Team · marcusgafka.com/#/robotics/frc-118

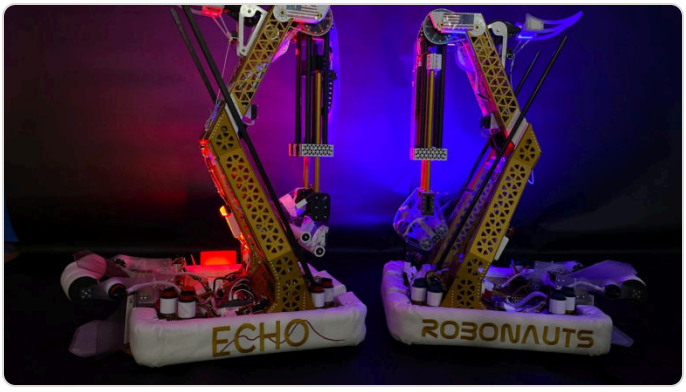
I spent four seasons with the Robonauts, going from opening CAD for the first time to designing entire subsystems and captaining the team.

CAD | Mechanical Design | Leadership | Drive Team

Growing up so close to NASA, I had the incredible opportunity in high school to apply to and join FRC Team 118, the Robonauts, a world-renowned team that has been competing for almost 30 years. On this team I grew my mechanical, robotics, and problem-solving skills, but most importantly, I realized that engineering was my calling.

I went from a quiet, shy, unconfident kid to a true leader. I opened CAD software for the first time on this team, and by the time I graduated I was designing entire subsystems with multiple degrees of freedom and complex geometry. In my senior year I was chosen as Team Captain and for the drive team, and the mentor I worked most closely with wrote my college recommendation letters.

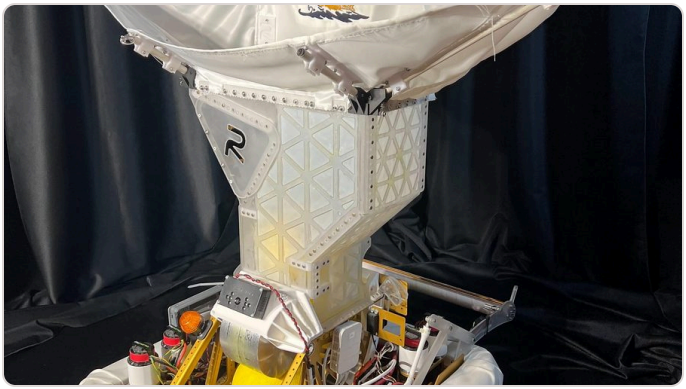
FRC and Team 118 were instrumental in my development as an engineer. I'm grateful to everyone involved, and I hope to pay it forward to the students I mentor now.



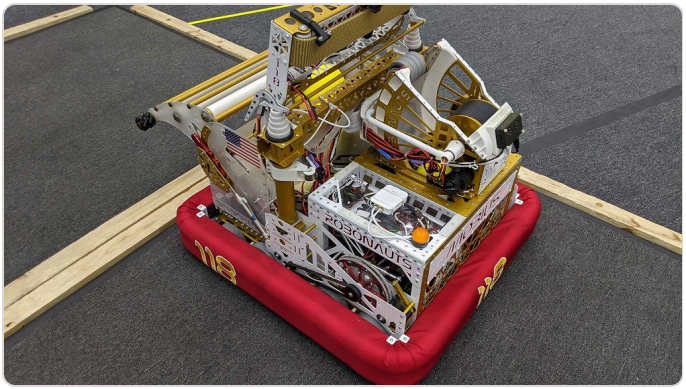
2023: Echo (photo: FRC 118 Robonauts)



2022: Horizon (photo: FRC 118 Robonauts)



2021: Marauder (photo: FRC 118 Robonauts)



2020: Mobius (photo: FRC 118 Robonauts)

LINKS

Robonauts Robots ↗